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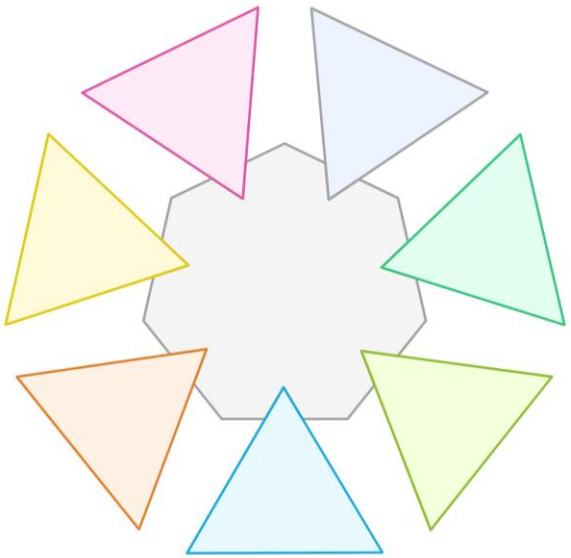
Smart In-Car Health Guardian

Automated Emergency Response System



Heart Beat Monitoring

Continuously tracks heart rate via steering wheel sensors

ML Anomaly Detection

Learns heart rate patterns to minimize false alarms

Contextual Verification

Cross-references data to confirm medical emergencies

GPS Integration

Initiates emergency response with driver cancellation

Medical Facility Routing

Identifies suitable hospitals within a radius

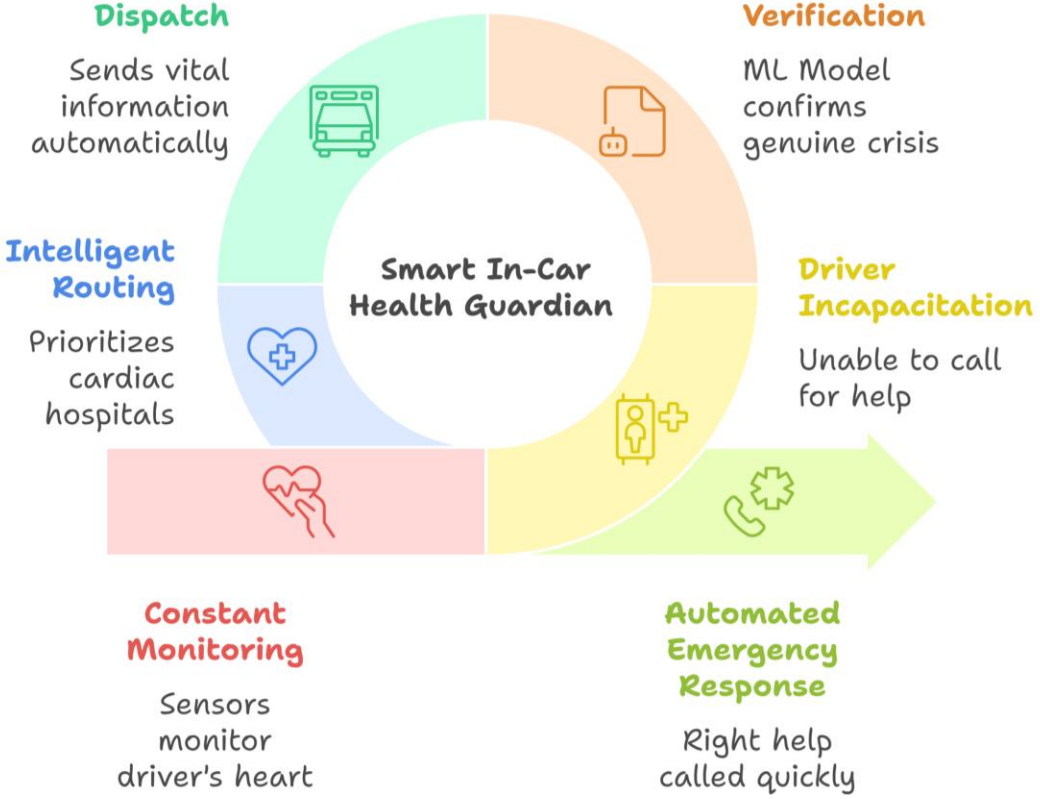
Multi-Channel Alerts

Transmits alerts to Hospitals, Patrolling Vehicles & Family Members

Crash Detection Override

Also triggers response upon airbag deployment for the stakeholders.

How does our Idea solves the problem?



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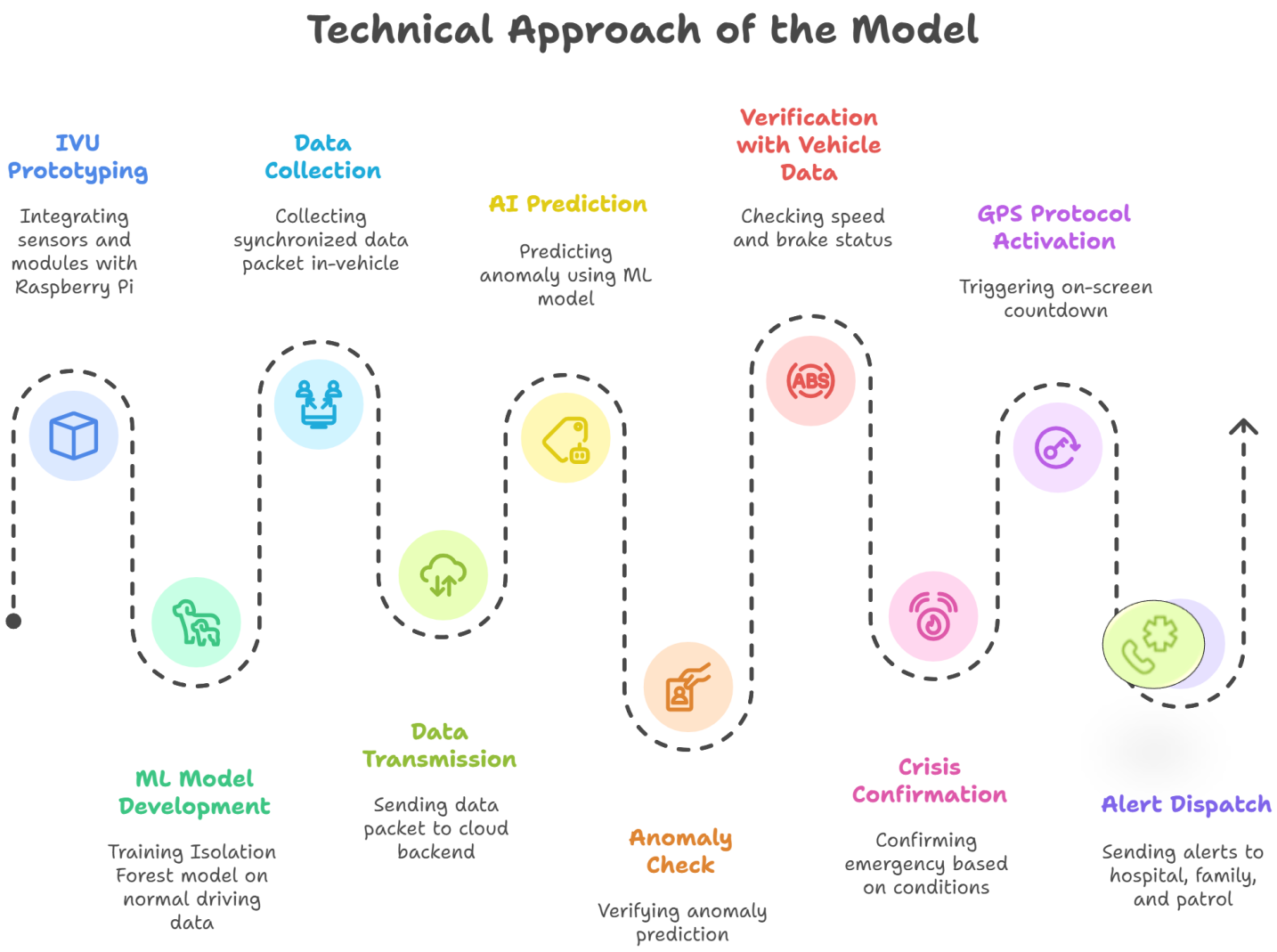
TECHNICAL APPROACH



- This system uses a Raspberry Pi 4 to process real-time heart rate data from a TCRTXXX sensor, applying LSTM Autoencoder and Decision Tree models to detect cardiac anomalies and employing k-d tree algorithms to instantly locate the nearest hospital or ambulance in an emergency.

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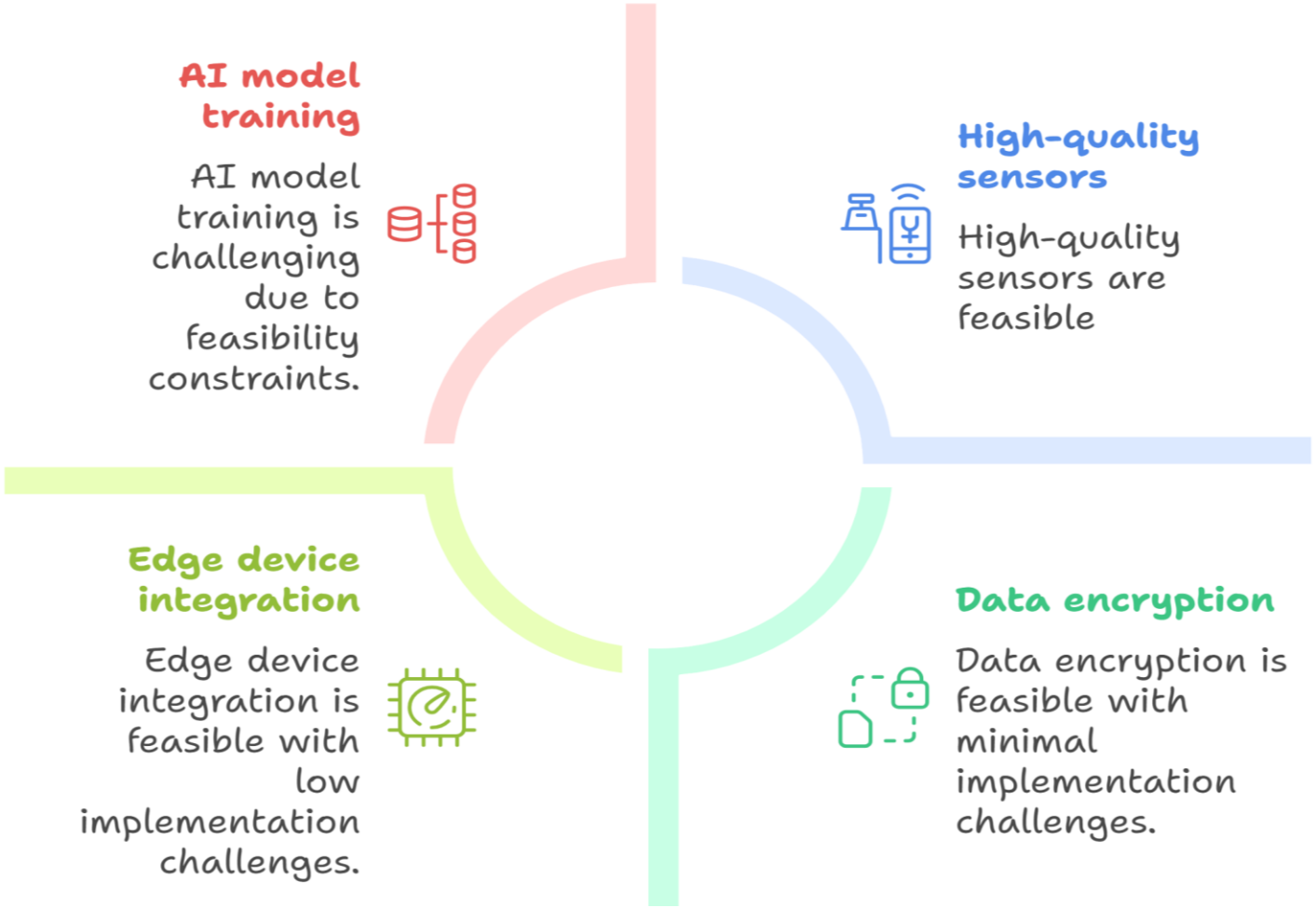
FEASIBILITY AND VIABILITY

Feasibility and Viability in Automotive Safety Systems



- This project is highly feasible due to easy sensor integration, efficient ML, and high market demand.
- Key challenges are ensuring sensor reliability with a loose grip and balancing accuracy to avoid false alarms while catching every real emergency.
- Solutions include using high-quality sensors, diverse ML training, data encryption, and an integrated, standardized emergency response system.

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IMPACT AND BENEFITS

- This technology offers true peace of mind for your family.
- If a medical emergency happens on the road, leaving a driver unable to call, it acts as their silent guardian.
- It instantly alerts responders with a precise location, turning a moment of helplessness into a story of survival and letting you worry less.

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RESEARCH AND REFERENCES

Details / Links of the reference and research work:

- https://docs.google.com/document/d/1OKEcv683cyiB9HHDHkJbd-X9zjG8TNkopHfk7uKvmY4/edit?usp=sharing
- https://drive.google.com/drive/folders/1Tr7jt_LT80KB32ZSodSzcr_Ke7tvACEm?usp=sharing